ABSTRACT

Medical image-based classification for rare brain tumors is a critical but challenging task due to the limited availability of quality data and overlapping visual features across different tumor types. This project proposes a deep learning-based solution named **RareDetect**, which employs a custom deep learning model to classify MRI brain images into four categories: glioma, meningioma, pituitary tumor, and no tumor.

The proposed model's performance is compared against two well-known pretrained models — **VGG16** and **ResNet50** — using evaluation metrics such as accuracy, precision, recall, and F1-score. The objective is to demonstrate that a tailored custom architecture can outperform standard CNNs in detecting underrepresented and complex medical cases. This system has the potential to serve as a decision-support tool for early diagnosis and medical research on rare brain diseases.

Keywords: CNN, VGG16, ResNet50, Brain Tumor, Glioma, Meningioma, MRI Classification, Custom Algorithm, Deep Learning, Rare Disease Detection

CHAPTER 1

INTRODUCTION

The rise of artificial intelligence (AI) and machine learning (ML) in the medical domain has opened new doors for faster, more accurate diagnosis of diseases using imaging data such as X-rays, CT scans, and MRIs. Among these, deep learning-based image classification has proven particularly powerful, with models capable of learning subtle patterns often missed by the human eye.

However, a major gap exists in the detection of rare diseases — especially in the field of neuroimaging, where MRI scans of brain tumors like glioma, meningioma, and pituitary tumors present complex and overlapping features. These diseases are less common, which leads to limited data, poor model generalization, and higher chances of misclassification.

To address this, the project titled **RareNet** proposes a robust solution using a custom deep learning algorithm tailored to detect and classify MRI brain scans into four key categories: *glioma*, *meningioma*, *pituitary tumor*, and *no tumor*.

The project will use open-source datasets and apply standard preprocessing techniques like resizing, normalization, and augmentation. A comparative study will be conducted between the proposed algorithm and established deep learning models — **VGG16** and **ResNet50** — using evaluation metrics such as accuracy, precision, recall, and F1-score.

The outcome will be a classification system that not only identifies tumors more accurately but also improves reliability for rare or underrepresented disease classes, supporting clinical diagnostics in real-world environments.

CHAPTER 2

OBJECTIVE

The primary aim of this project is to develop a deep learning-based system for accurate detection and classification of rare brain tumors using MRI data. The specific objectives of the project are as follows:

1. To design and implement a custom image classification algorithm for rare brain tumor detection using MRI scans.
2. To utilize public medical image datasets and preprocess them for optimal training.
3. To train and evaluate well-known pretrained models (**VGG16** and **ResNet50**) for comparison.
4. To evaluate and compare model performance using metrics such as accuracy, precision, recall, and F1-score; and to visualize results in the form of graphs, comparison tables, and performance curves.
5. To analyze the advantages and limitations of each approach, with emphasis on rare class detection.
6. To analyze how the custom algorithm can improve reliability in rare case detection.
7. To support early-stage diagnosis in medical systems by improving classification reliability.

CHAPTER 4

LITERATURE SURVEY

1. **MRI-Based Brain Tumor Classification Using CNN** – A study by Patel et al. (2022): This study proposed an automated CNN-based model for classifying three types of brain tumors: glioma, meningioma, and pituitary. The authors emphasized the importance of image preprocessing techniques like resizing and normalization. Their results proved that deep CNNs can outperform traditional ML methods when working with spatial features in MRI images.[1]

2. **Binary Brain Tumor Detection Using Simplified CNN** – A study by Dandale et al. (2021): This research focused on binary classification of brain tumors (tumor vs no tumor) using a lightweight CNN architecture. Despite its simplicity, the model achieved reliable accuracy and proved useful for small datasets. The study also stressed the importance of kernel size and dropout layers for controlling overfitting.[2]

3. **Transfer Learning-Based Multiclass Tumor Classification** – A study by Mirza et al. (2021): In this paper, the authors used transfer learning with models like VGG16, ResNet50, and InceptionV3 for multiclass tumor classification. They demonstrated how pretrained models, when fine-tuned on medical image datasets, offer high accuracy with minimal training time. This supports our approach of model comparison in RareNet.[3]

4. **CNN for Improved Sensitivity in Low-Sample Tumor Classes** – A study by Sharma et al. (2020): The researchers designed a CNN architecture optimized for detecting rare tumor classes with very limited training samples. The study highlighted how domain-specific customizations to CNN layers can drastically improve sensitivity and specificity — especially in medical imaging tasks.[4]

5. **Ensemble-Based CNN for Robust Medical Image Classification** – A study by Yadav et al. (2023): This study presented an ensemble method combining multiple CNN architectures to improve classification robustness. Although computationally intensive, the approach achieved state-of-the-art performance.[5]

CHAPTER 5

PROBLEM STATEMENT

Rare brain tumors such as glioma, meningioma, and pituitary tumors often go undetected due to limited diagnostic data and lack of expert accessibility, especially in rural areas. Manual interpretation of MRI scans is time-consuming and subject to human error.

This project aims to develop an AI-based classification system using a custom deep learning model. The system will classify brain MRI images into four categories: Glioma, Meningioma, Pituitary Tumor, and No Tumor, and its performance will be compared with standard models like VGG16 and ResNet50 to evaluate improvements in accuracy and reliability.

CHAPTER 6

CONCLUSION

The RareDetect project presents a focused and impactful solution to one of the most pressing challenges in modern healthcare — the early detection of rare brain tumors. By applying deep learning to MRI image classification, the system seeks to enhance diagnostic accuracy while reducing dependency on manual interpretation.

The development of a custom AI model, along with comparative analysis against standard architectures like VGG16 and ResNet50, provides both technical insight and practical value. The model is expected to perform efficiently across various tumor types, particularly those that are rare or subtle, making it a strong candidate for real-world clinical support systems.

If successful, this system can be adapted for other rare disease applications and integrated into telemedicine platforms, particularly in underserved or remote areas. The project not only contributes to the AI-in-healthcare domain but also reflects a step toward equitable, accessible, and technology-driven diagnosis for all.

CHAPTER 7

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